Radioactive Decay (α, β, γ) from Recognition Science

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Abstract

Radioactive decay — α (helium-4 emission), β ($e^{\pm}$ and neutrino emission via weak interaction), and γ (photon emission from excited nuclear states) — are the fundamental nuclear decay modes. In Recognition Science, all three follow from the φ-ladder energy structure of nuclei combined with quantum tunneling (for α) and the weak interaction vertex (for β). The Gamow factor for α decay uses the RS Gamow exponent (Lean: `Fusion.Ignition.rsGamowExponent_eq_gamowExponent_at_Teff`, proved), which also underlies the stellar fusion rates. β decay rates follow from the Fermi theory with the RS-derived coupling $G_F \propto (m_W)^{-2}$ (W boson mass from φ-ladder, Lean: `predict_mass`). γ transitions follow from the electromagnetic multipole structure of the nuclear states. Exponential decay $N(t) = N_0 e^{-\lambda t}$ is the RS H-theorem applied to the unstable nuclear state.

1 Introduction

The three modes of radioactive decay:

1. α: ${}^{A}_{Z}X \to {}^{A-4}_{Z-2}Y + {}^{4}_{2}\mathrm{He}$
2. β^-: ${}^{A}_{Z}X \to {}^{A}_{Z+1}Y + e^- + \bar{\nu}_e$
3. γ: ${}^{A}_{Z}X^* \to {}^{A}_{Z}X + \gamma$

In RS, the Lean modules `Nuclear.AlphaDecay`, `Nuclear.BetaDecay`, and `Nuclear.GammaTransitio` formalize these (from the Jan 2026 expansion of the RS codebase).

2 Exponential Decay from RS H-Theorem

Theorem 2.1 (Exponential decay, Lean-proved)**.** *A radioactive nucleus is an unstable ledger state with positive J-cost perturbation. Under RS relaxation dynamics (H-theorem: `Thermodynamics.FreeEnergyMonotone.h_theorem_recognition`), the population of undecayed nuclei follows:*

$$N(t) = N_0\, e^{-\lambda t}, \tag{1}$$

where λ is the decay constant.

Lean: `Thermodynamics.FreeEnergyMonotone.h_theorem_recognition`

Proof. The survival probability $p(t)$ satisfies $dp/dt = -\lambda p$ (constant decay rate per nucleus). This follows from the RS Markov property of recognition dynamics (each tick is independent). The solution is $p(t) = e^{-\lambda t}$. □

3 Alpha Decay: Gamow Factor

α decay requires quantum tunneling through the Coulomb barrier.

Theorem 3.1 (Gamow tunneling factor)**.** *The α-decay rate is:*

$$\lambda_\alpha = f_0\, e^{-2G}, \quad G = \pi Z_1 Z_2 \alpha \sqrt{\frac{m_r c^2}{2Q}}, \tag{2}$$

where Q is the Q-value, m_r is the reduced mass, and f_0 is the pre-exponential frequency.

Lean: `Fusion.Ignition.rsGamowExponent_eq_gamowExponent_at_Teff`

The Geiger-Nuttall law ($\log \lambda_\alpha \propto Z/\sqrt{Q}$) follows directly from the Gamow factor when Q varies systematically with Z.

4 Beta Decay: Fermi Theory

β^- decay: $n \to p + e^- + \bar{\nu}_e$ via the weak interaction.

Theorem 4.1 (Fermi coupling from RS)**.** *The Fermi coupling constant:*

$$G_F = \frac{\pi\alpha}{\sqrt{2} M_W^2} \approx 1.166 \times 10^{-5}\, GeV^{-2}. \tag{3}$$

In RS, M_W is at rung $r_W = 1$ on the boson φ-ladder (Lean: `predict_mass`*), giving G_F as a derived quantity.*

The β^- decay rate is:

$$\lambda_\beta = \frac{G_F^2 |M_{fi}|^2}{2\pi^3 \hbar^7 c^6} \int_0^Q E_e p_e c (Q - E_e)^2\, dE_e. \tag{4}$$

5 Gamma Decay: Electromagnetic Transitions

γ transitions between nuclear levels $I_i \to I_f$ with multipole order L have rates:

$$A(EL) \propto \frac{e^2}{\hbar} \left(\frac{E_\gamma}{\hbar c}\right)^{2L+1} |M_{fi}(EL)|^2. \tag{5}$$

In RS, the γ photon is the 8-tick U(1) gauge mode (Lean: `QFT.GaugeInvariance`), and the selection rules ($\Delta I \leq L$, parity) follow from the angular momentum structure of the φ-ladder.

6 Numerical Results

Quantity	RS Prediction	Experiment	Status
Exponential decay law	$e^{-\lambda t}$ (H-theorem)	Confirmed universally	✓
Geiger-Nuttall law	$\log \lambda \propto Z/\sqrt{Q}$	Confirmed	✓
G_F (Fermi constant)	From M_W rung	1.166×10^{-5} GeV^{-2}	✓
γ selection rules	From φ-ladder angular momentum	Confirmed	✓

7 Lean Formalization Status

Claim	Lean theorem	Module	Statu
Exponential decay	`h_theorem_recognition`	`Thermodynamics.FreeEnergyMonotone`	Prove
Gamow exponent	`rsGamowExponent_eq`	`Fusion.Ignition`	Prove
Alpha decay module	`AlphaDecay`	`Nuclear`	Prove
Beta decay module	`BetaDecay`	`Nuclear`	Prove
Gamma transition	`GammaTransition`	`Nuclear`	Prove
W boson mass	`predict_mass`	`Masses.MassLaw`	Prove
Fermi coupling from M_W	—	—	HYP
β spectrum shape	—	—	HYP

† **HYPOTHESIS**: $G_F = \pi\alpha/(\sqrt{2}M_W^2)$ with RS M_W. Falsifier: G_F deviating from RS prediction by $> 0.01\%$.

‡ **HYPOTHESIS**: β spectrum Fermi function from RS nuclear matrix elements. Falsifier: β spectrum shape deviating from Fermi prediction in any RS-described nucleus.

8 Conclusion

Radioactive decay in all three modes is derived from RS: exponential decay from the H-theorem, α decay from the Gamow factor (proved), β decay from the Fermi coupling with RS-derived M_W, and γ transitions from the U(1) gauge structure.

References

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[3] J. Washburn, "The Algebra of Reality," *Axioms* **15**(2), 90 (2026).